

Beyond In-Context Learning on LLMs: Rethinking Few-Shot Adaptation for Structured Tasks

Sunita Sarawagi
IIT Bombay
sunita@iitb.ac.in

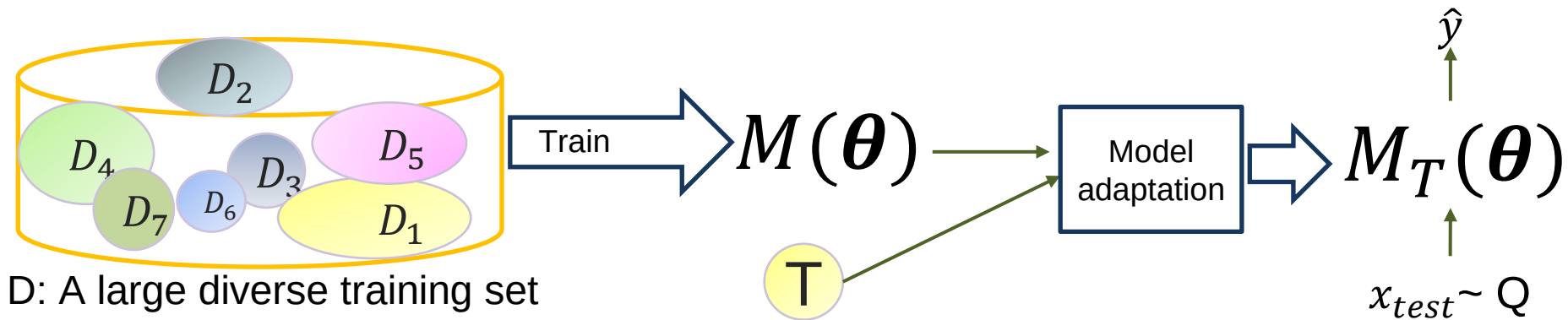
Few-shot Model Adaptation

Given a trained model $M(\theta)$ on a large diverse dataset D

User interested in a target task Q , gives a **small** labelled data

$$T: \{x_1 \ y_1 \ x_2 \ y_2 \ x_3 \ y_3 \ \dots \ x_k \ y_k\} \sim Q$$

Model adaptation = Find the best blend of $M(\theta)$ and T



D : A large diverse training set

D may contain a part D_j that is relevant to target task, but it is not explicitly identified.

Example: model adaptation

$M(\theta)$

Target task

Conversation assistant



→ New users



Text-to-SQL system



Concerts



Football



Travel



Countries



Books

→ Private pharma database

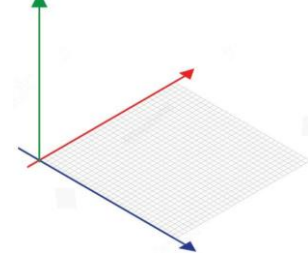


Foundation model for time series → Forecast for new org

Trained on several system logs

Need to adapt to new covariates

The challenges of model adaptation



- Accuracy: requires delicate balance.

T : Small but relevant

$M(\theta)$: Large, mixed relevance



Adapt too much $\rightarrow$ overfit, forget old

Adapt too little $\rightarrow$ No change

- Robustness: Small T leads to high variance across $T \sim Q$
 - Cannot introduce additional instabilities, e.g. due to ordering
- Efficiency: Θ is large
 - Time of adaptation: Offline Vs Online (On-the-fly)
 - Test time: time for using after adaptation.

Model adaptation over the Ages

Thousands of paper under variants like domain adaptation, transfer-learning, few-shot learning..

Adaptation of Maximum Entropy Capitalizer: Little Data Can Help a Lot

Ciprian Chelba · A. Acero · Computer Science · [Computer Speech and Language](#) · 2006

Domain adaptation of natural language processing systems

Fernando C Pereira · John Blitzer · Computer Science, Linguistics · 2008

Domain adaptation of information extraction models

Rahul Gupta · Sunita Sarawagi · Computer Science · SGMD · 20 March 2009

Adversarial Discriminative Domain Adaptation

Eric Tzeng · Judy Hoffman · Kate Saenko · Trevor Darrell

[Computer Vision and Pattern Recognition](#) · 17 February 2017

Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks

Chelsea Finn · P. Abbeel · S. Levine · Computer Science · [International Conference on Machine Learning](#) ·

Deep Transfer Learning with Joint Adaptation Networks

Mingsheng Long · Hanhua Zhu · Jianmin Wang · Michael I. Jordan

[International Conference on Machine Learning](#) · 21 May 2016

Domain Adaptation of Conditional Probability Models Via Feature Subsetting

Sandeepkumar Satpal · Sunita Sarawagi · Computer Science ·

[European Conference on Principles of Data Mining...](#) · 17 September 2007

Structured Case-based Reasoning for Inference-time Adaptation of Text-to-SQL parsers

Abhijeet Awasthi · Soumen Chakrabarti · Sunita Sarawagi · Computer Science ·

[AAAI Conference on Artificial Intelligence](#) · 10 January 2023

LoRA: Low-Rank Adaptation of Large Language Models

J. E. Hu · Yelong Shen · +4 authors · Weizhu Chen · Computer Science

[International Conference on Learning...](#) · 17 June 2021

Unsupervised Domain Adaptation by Backpropagation

Yaroslav Ganin · V. Lempitsky · Computer Science · [International Conference on Machine Learning](#) ·

26 September 2014

Model adaptation methods

- Fine-tuning and its variants
- Mixture of experts
- Task-vectors
- Matching-based methods

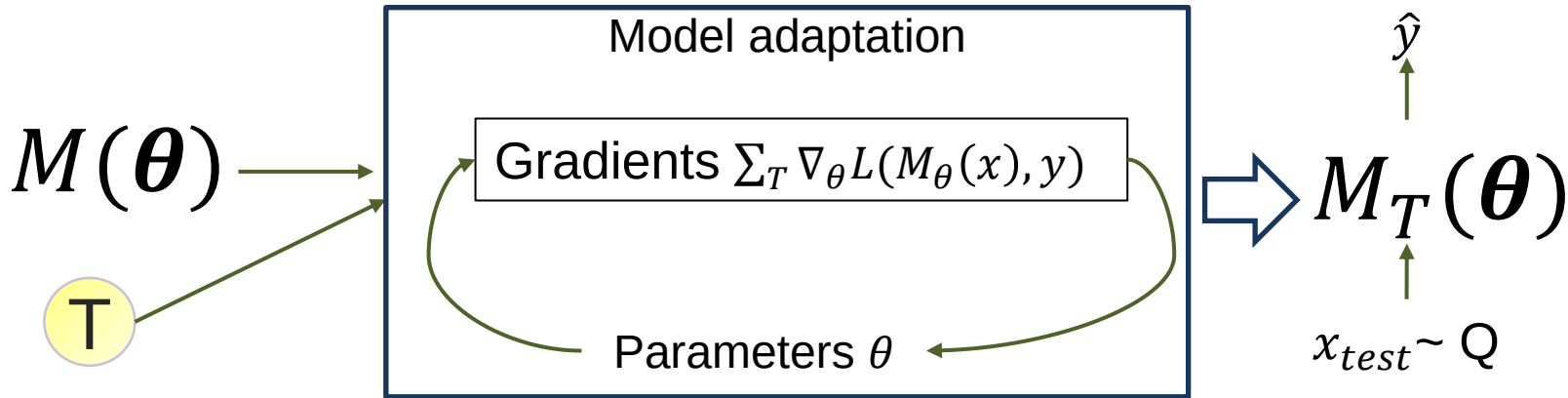
Each category explicitly planned for adaptation either when

- Training $M(\Theta)$, or when
- Adapting with T

Fine-tune parameters via gradient descent on T

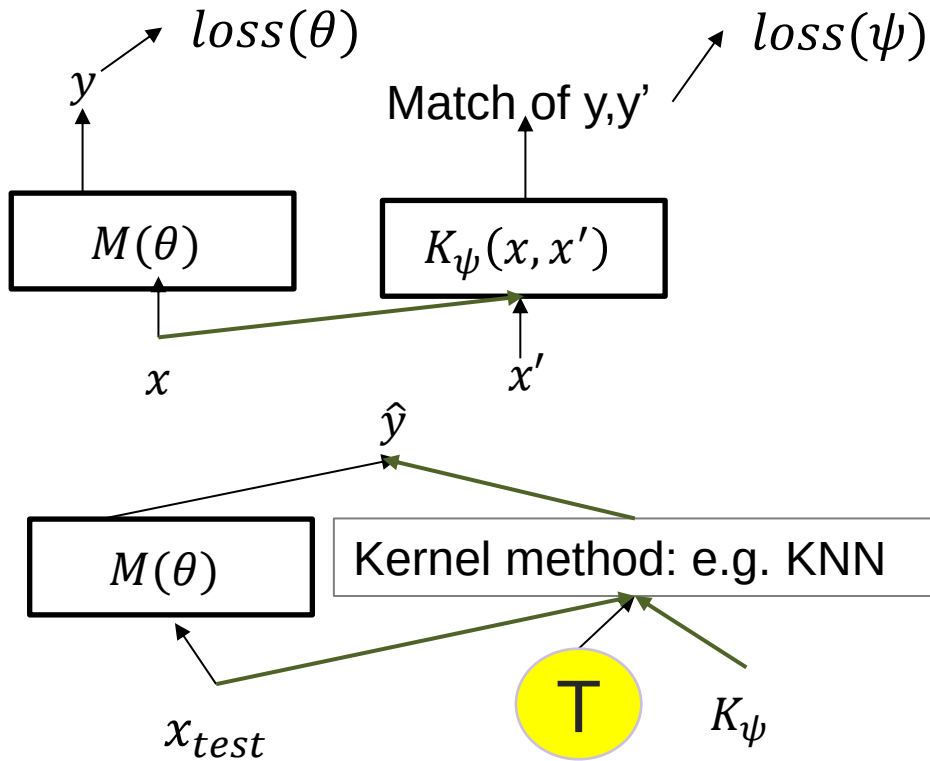
Training of $M(\theta)$:
Normal loss on D

Adaptation with T:
Update θ via gradient descent on T



Explicit update of model parameters Θ to minimize loss on target data T
Generally provides good accuracy for modest sized T

Adaptation with Matching Networks



Training:

- Train Θ as usual
- Contrastive $K_\psi(x, x')$ so example pairs with same labels are close

Adaptation:

- Use kernel method to get predictions from T using $K_\psi(x, x'), x' \in T$.
- Blend with predictions from $M(x|\theta)$

- Labeled Memory Networks for Online Model Adaptation. In AAAI, 2018. Shiv Shankar and Sunita Sarawagi
- Speech-enriched Memory for Inference-time Adaptation of ASR Models to Word Dictionaries Mittal, Sarawagi, Jyothi. EMNLP 2023.
- Conditional Tree Matching for Inference-Time Adaptation of Tree Prediction Models. Varma, Awasthi, Sarawagi ICML 2023

An example matching-based adaptation for Text2SQL

Text to SQL

Schema

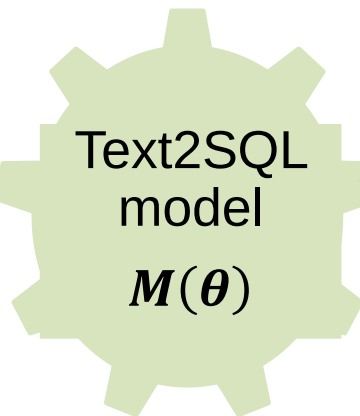
Department		
<u>Id</u>	Name	Description
14	Informatik	..
11	Software system	..
21	Engineer	...

Degree programs		
<u>Id</u>	Name	Dept id
1	PhD	14
2	Masters	21
3	Bachelors	21

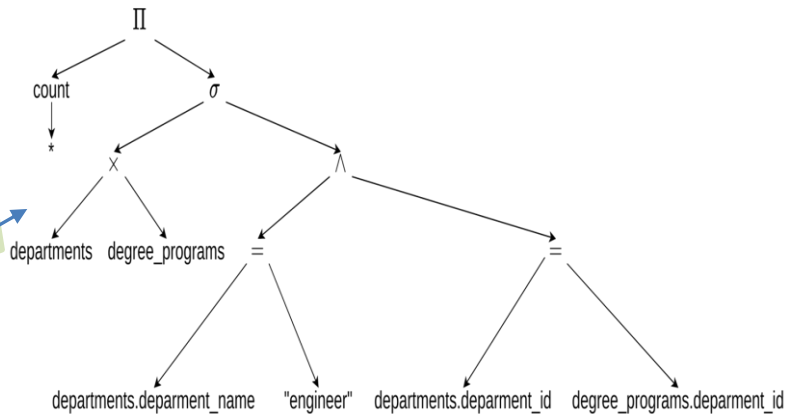
Question

How many degrees does the engineering department offer?

x



Relational Algebra Tree representation of SQL



y

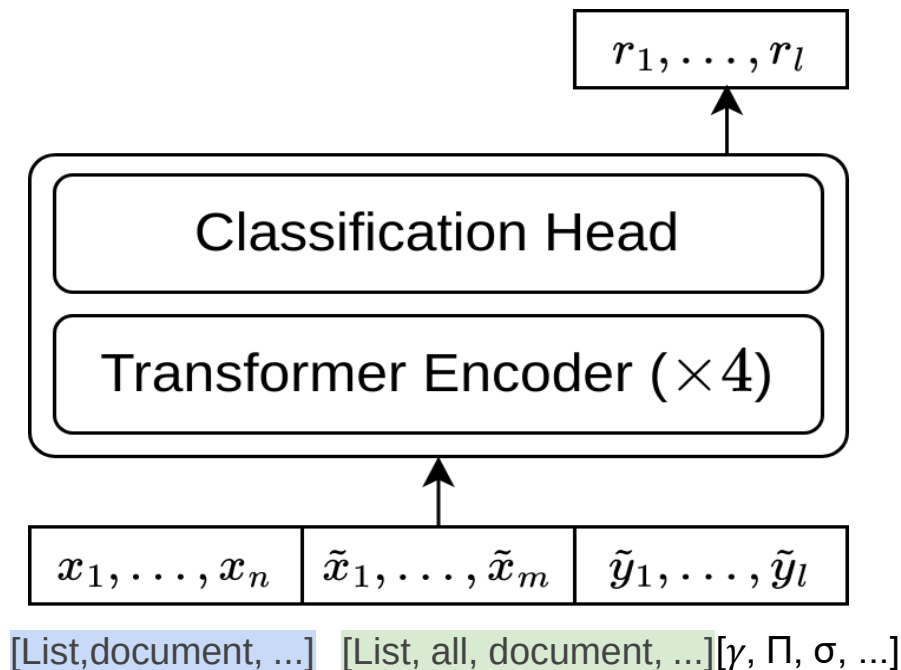
Designing matching kernel: $K((x, y), (x', y'))$

Two parts since y is structured.

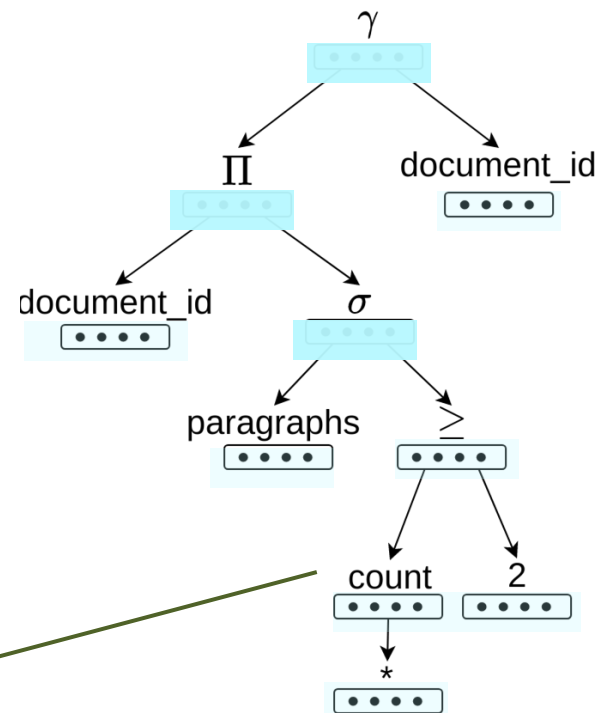
- Node-level reasoning of the relevance of specific subtrees of labelled examples to current question
 - Fine-grained sharing.
- Similarity based on alignment of respective trees → structurally informed

Fine-grained relevance scoring

Relevance score each subtree



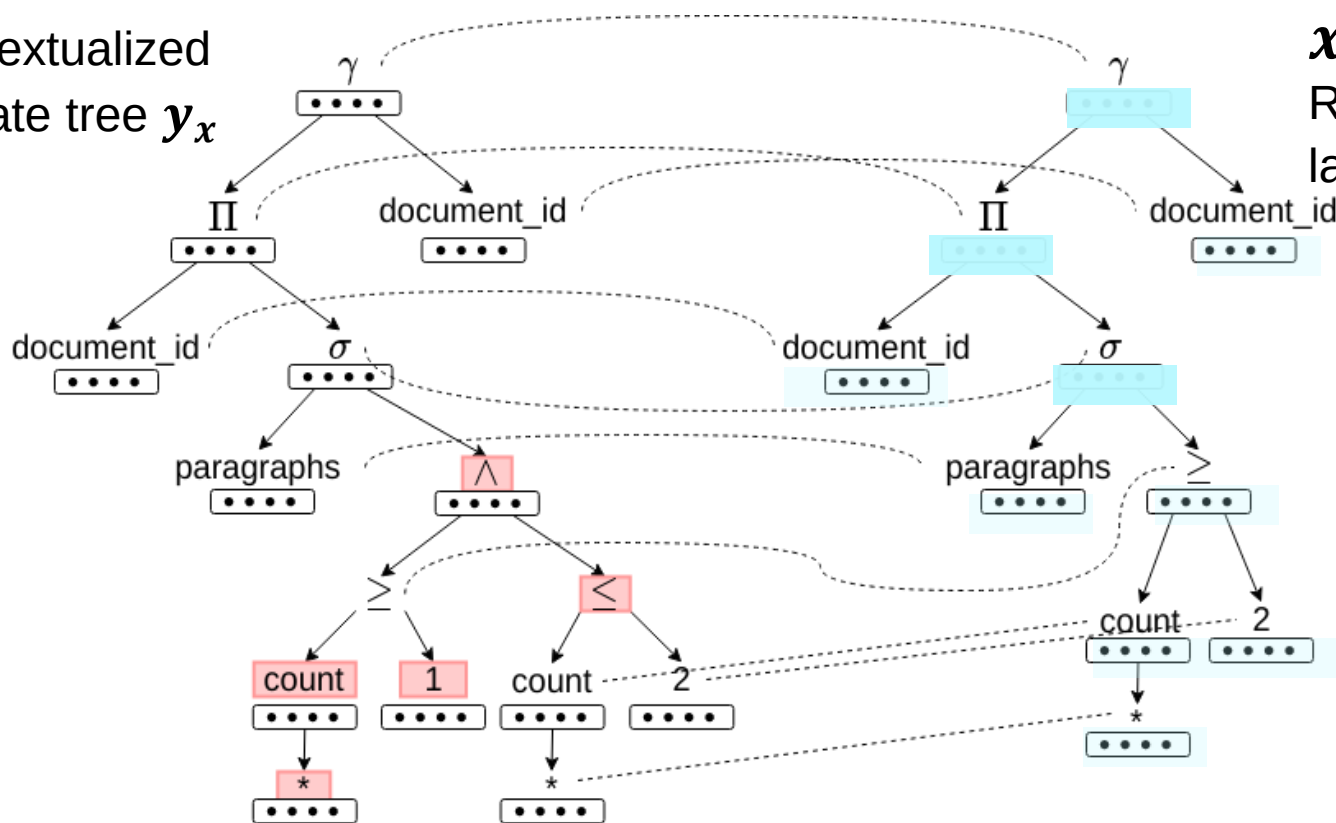
Wrong nodes marked irrelevant



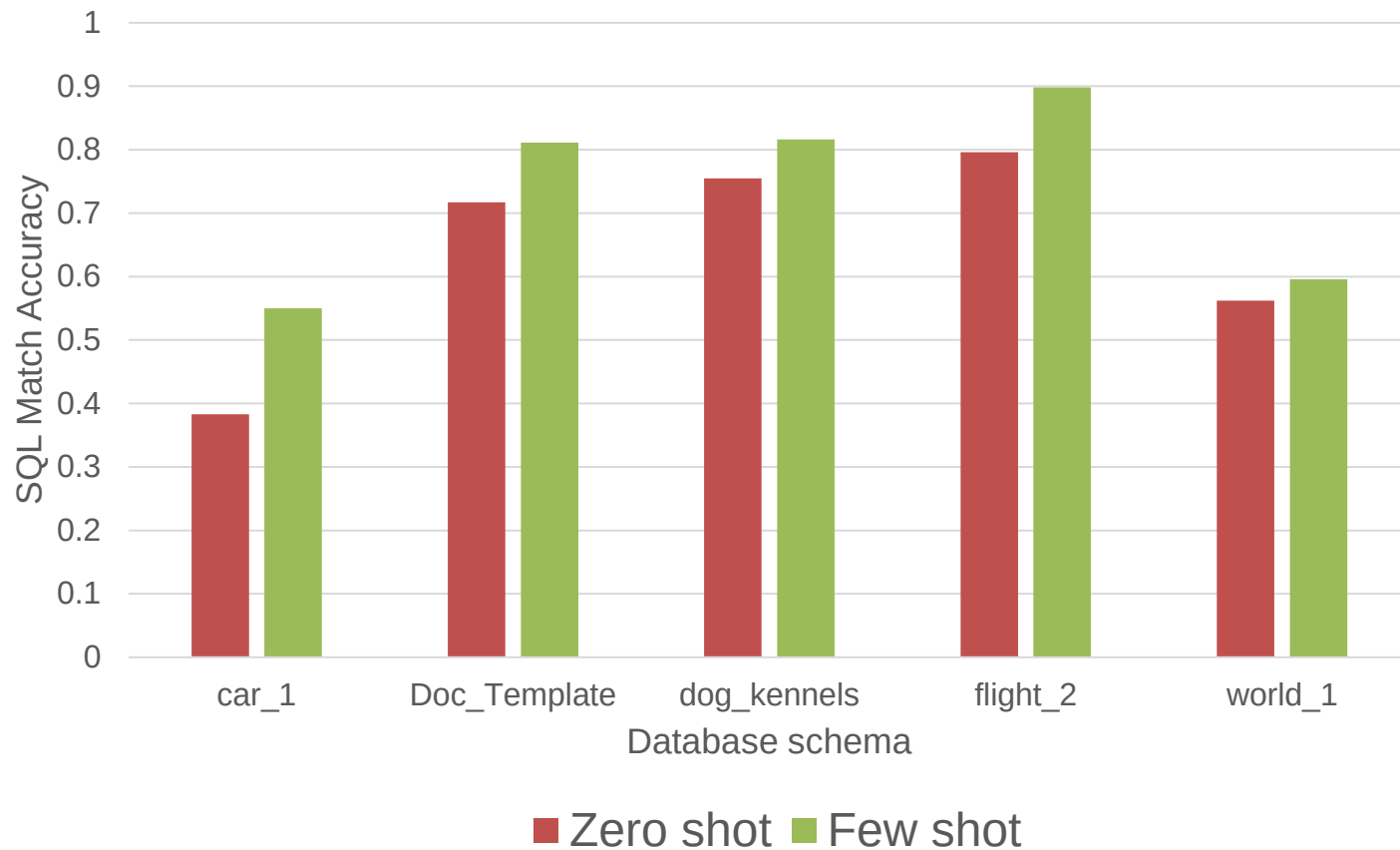
Alignment based relevance-weighted $K((x, y), (x', y'))$

x Contextualized
candidate tree y_x

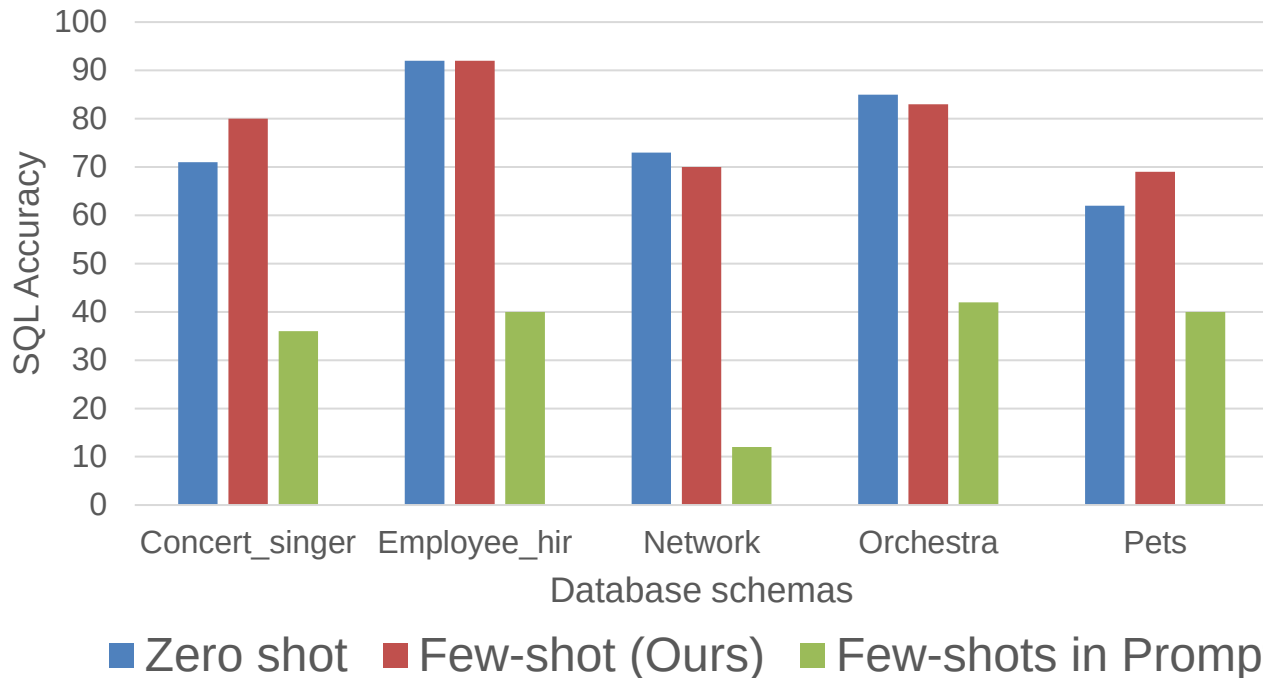
x' Contextualized
Relevance weighted
labeled tree $y'_{x'}$



Accuracy after adaptation with few examples



Greater robustness to irrelevant examples



Reasoning about fine-grained relevance of few-shots is important for robustness

Summary of previous adaptation approaches

All methods carefully designed for adaptation with T, very often with special networks and special losses during training, and/or during deployment

Enter the LLMs...

- Solution to model adaptation just emerges in the form of in-context learning!
- Models trained with next-token prediction loss on huge natural document collections can be adapted just with few labelled examples in-context..

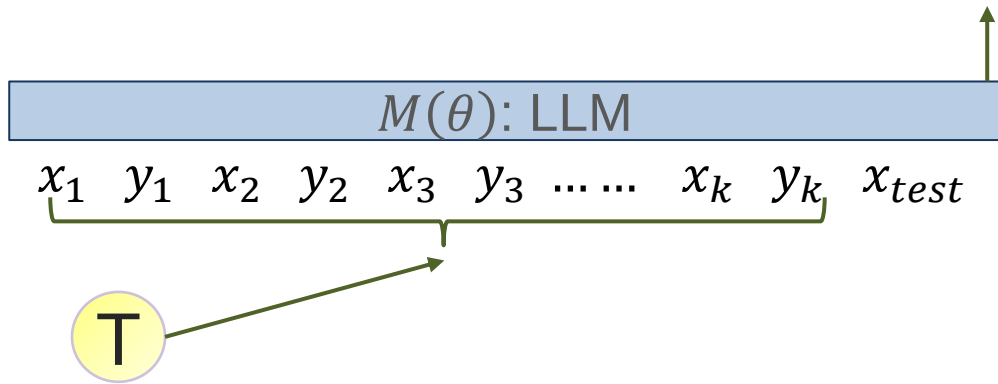


Adaptation via In-Context Learning in LLMs

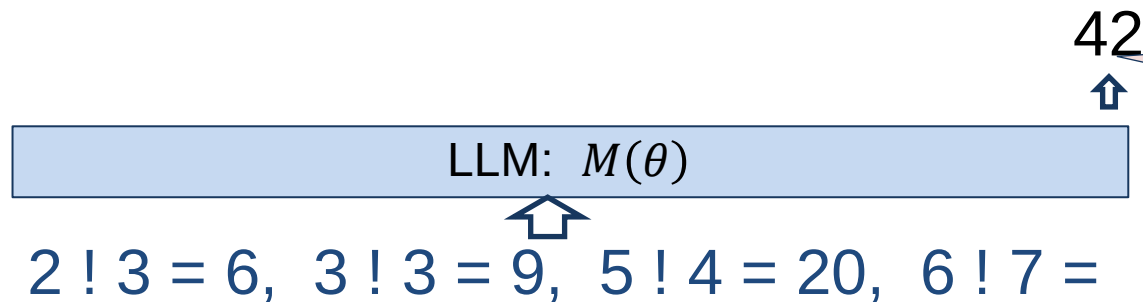
Training: Default

Adaptation: Just a forward pass.

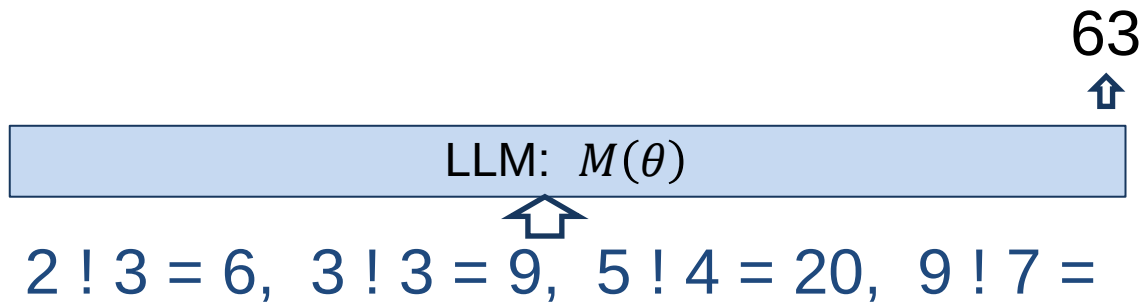
Predicted label $\hat{y} \sim M_T(x_{test}|\theta)$



Examples



Hardly surprising, if 42 is “the answer to life, universe and everything”



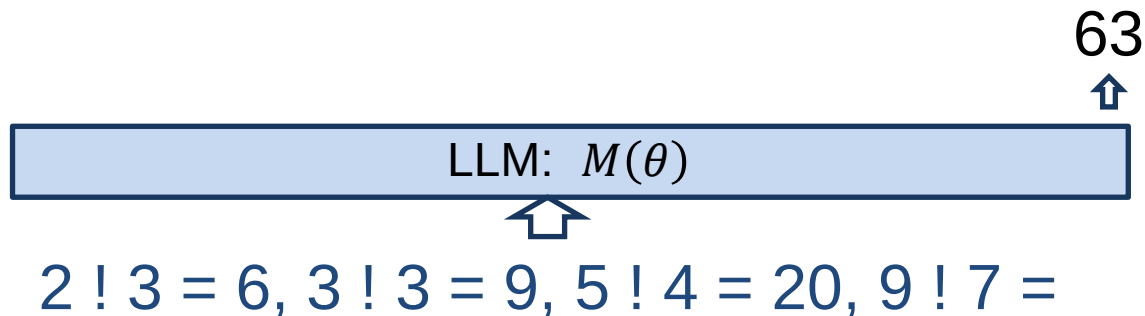
Why does ICL work?

- Hypothesis 1: Transformers implement gradient descent algorithm over IC examples
- Hypothesis 2: IC examples recognize tasks in pre-training e.g. via task vectors
- Hypothesis 3: Self-attention implements matching-based adaptation via induction heads

Hypothesis 2:

ICL identifies task from the pre-training set

ICL does task selection



LLMs provide the correct answer by recognizing this as a multiplication problem. They do not really learn to do multiplication from these examples.

How does task selection capability emerge in LLMs after pre-training with next token loss?

ICL does Task Selection

Pretraining documents:
mix of related topics



Coffee

Coffee is a popular drink ...

Coffee is a drink that invigorates

Coffee is a brewed drink made ...

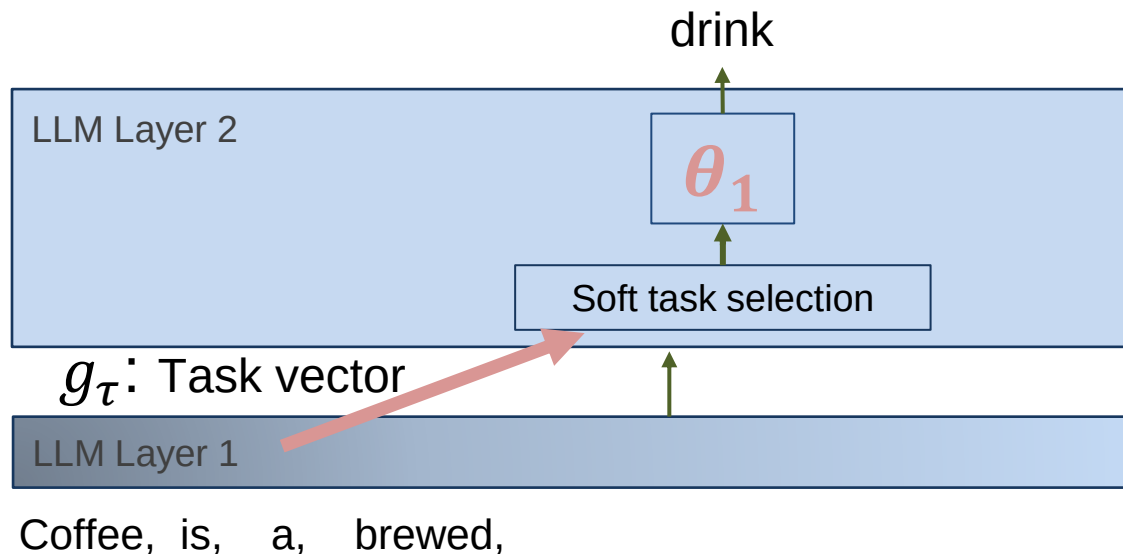


Music

The most popular music genre ...

The first electric guitar was ...

Prefix of sentence recognizes topic as latent vectors $g_\tau \rightarrow$ soft select task $\rightarrow$ recall words from related documents seen earlier.



- An explanation of in-context learning as implicit Bayesian inference. ICLR 2022. Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma
- The learnability of in-context learning. NeurIPS 2023. Noam Wies, Yoav Levine, and Amnon Shashua

ICL goes beyond task selection

Large LLMs do learn novel input-output mappings

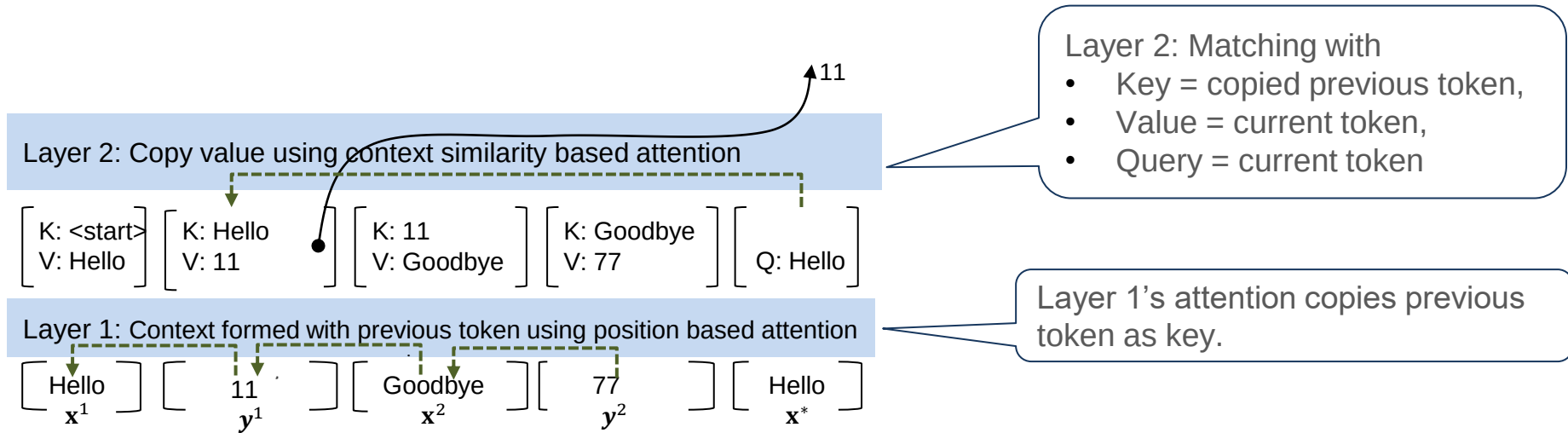
- Replacing positive/negative with foo/bar or flipping labels works for sentiment classification
- Unseen key-value associative mappings are learned
- Totally synthetic formal Markovian languages are learned

- What do language models learn in context? the structured task hypothesis. ACL 2024. Jiaoda Li, Yifan Hou, Mrinmaya Sachan, and Ryan Cotterell.
- Why larger language models do in-context learning differently? ICLR 2024. Zhenmei Shi, Junyi Wei, Zhuoyan Xu, and Yingyu Liang

Hypothesis 3: Self-attention implements matching-based adaptation

ICL= Matching with Induction circuits

Induction circuits created over two layers

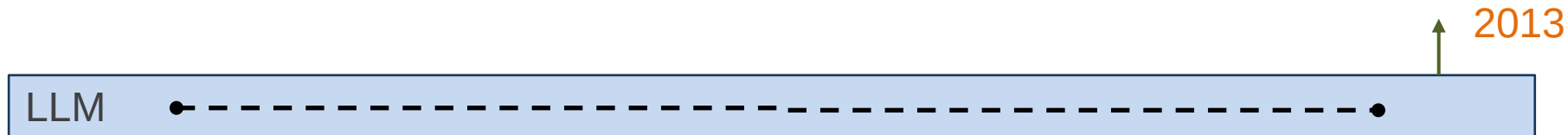


But why do transformer parameters orient to create in-context induction heads during next-token loss pre-training?

Pre-training data contains repeated phrases

Corpus contains co-occurring phrases in similar templates.

care. We will *estimate* infant and neonatal mortality rates again in *2015* to see if this trend continues and, if so, to assess how it *can be* reversed. Infant mortality in *2013* was

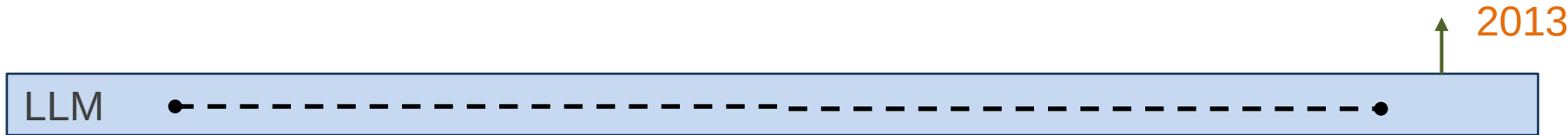


.. neonatal mortality rates again in 2015 to see if thisinfant mortality in

Pre-training data contains repeated phrases

Destroy repetitions from
context during training
causes ICL to drop by 50%
as against 2% random drops

care. We will **estimate** infant and neonatal ~~mortality rates~~
~~again in 2015~~ to see if this trend continues and, if so, to
assess how it **can be** reversed. Infant mortality in **2013** was



.. strawberries ran December crazy to see if thisinfant mortality in

Random tokens

How far are we in our understanding of ICL?

- My bet: For NLP, ICL most likely a blend of task selection and matching with induction heads
- Our understanding is still incomplete
 - Most studies on scalar outputs: classification and regression
 - What about structured outputs for example, Sequence to Sequence learning?
 - Most studies on small models
 - What about very large models?

Evaluating ICL in practice

Main conclusions

- Modern LLMs can learn new classification tasks and new regular languages via ICL
 - Larger models more capable of flipping old learning
- On long context LLMs, ICL competitive with fine-tuning but struggles on difficult tasks with large label spaces
- Out of distribution generalization was earlier thought to be better for ICL but there are some conflicting studies...verdict still out.

(small T)

Fine-tuning

ICL

Accuracy



Robustness



Efficiency

Online

Adapt



Test



Forgetting old tasks



Others



Ease of use in multi-tenancy models

Evaluating ICL on Structured Sequence to Sequence Tasks

Structured Sequence to Sequence learning

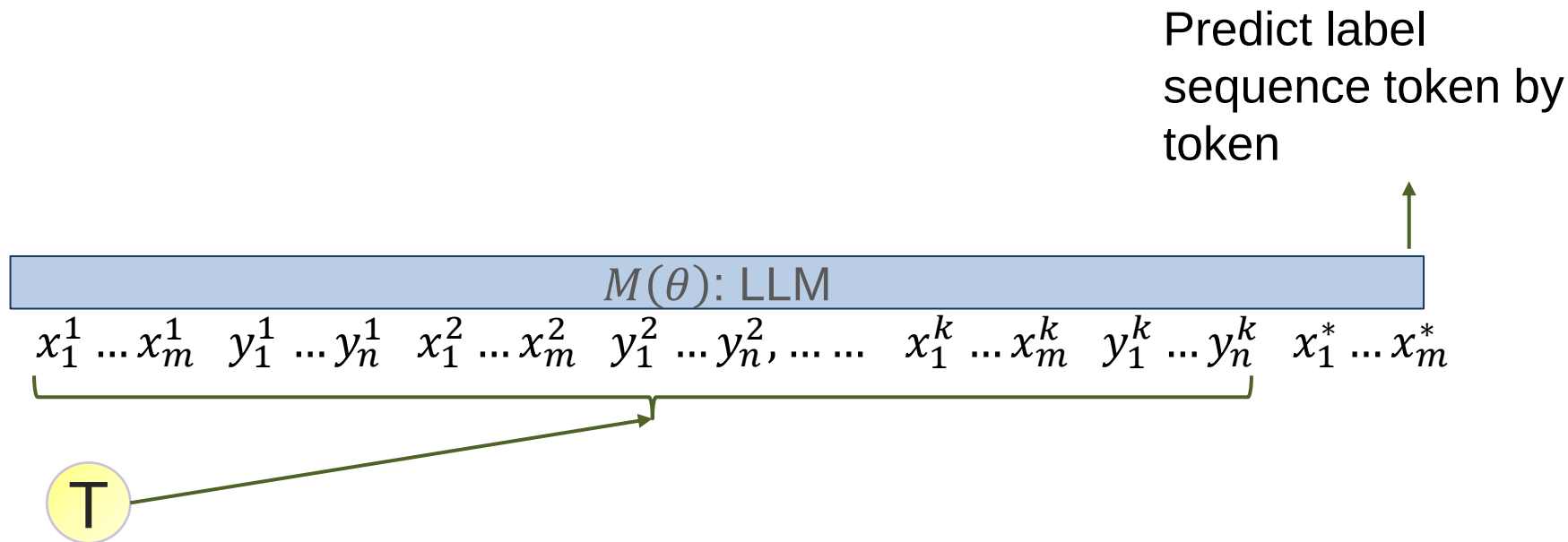


$P(\mathbf{y}|\mathbf{x})$ can be decomposed in terms of simpler structural units

Resource poor applications that need few-shot adaptation

- Semantic parsing for custom APIs/languages
- Translation from or to rare languages
- Text2SQL for new private enterprise schema

ICL on Sequence to Sequence tasks



Structured $\rightarrow$ true label $y_1^* \dots, y_n^*$ to be composed out of **sub-parts** of multiple IC examples.

No single example is wholly relevant $\rightarrow$ Irrelevant parts interfere!

Evaluating ICL on Seq2seq tasks



Testing on synthetic formal languages to ensure that dataset is leaked in large LLMs

Synthetic generator inspired by classical statistical lexical translation models

Brown, P. F., Della Pietra, S. A., Della Pietra, V. J., and Mercer, R. L. The mathematics of statistical machine translation: Parameter estimation. *Computational Linguistics*, 19(2):263–311, 1993.

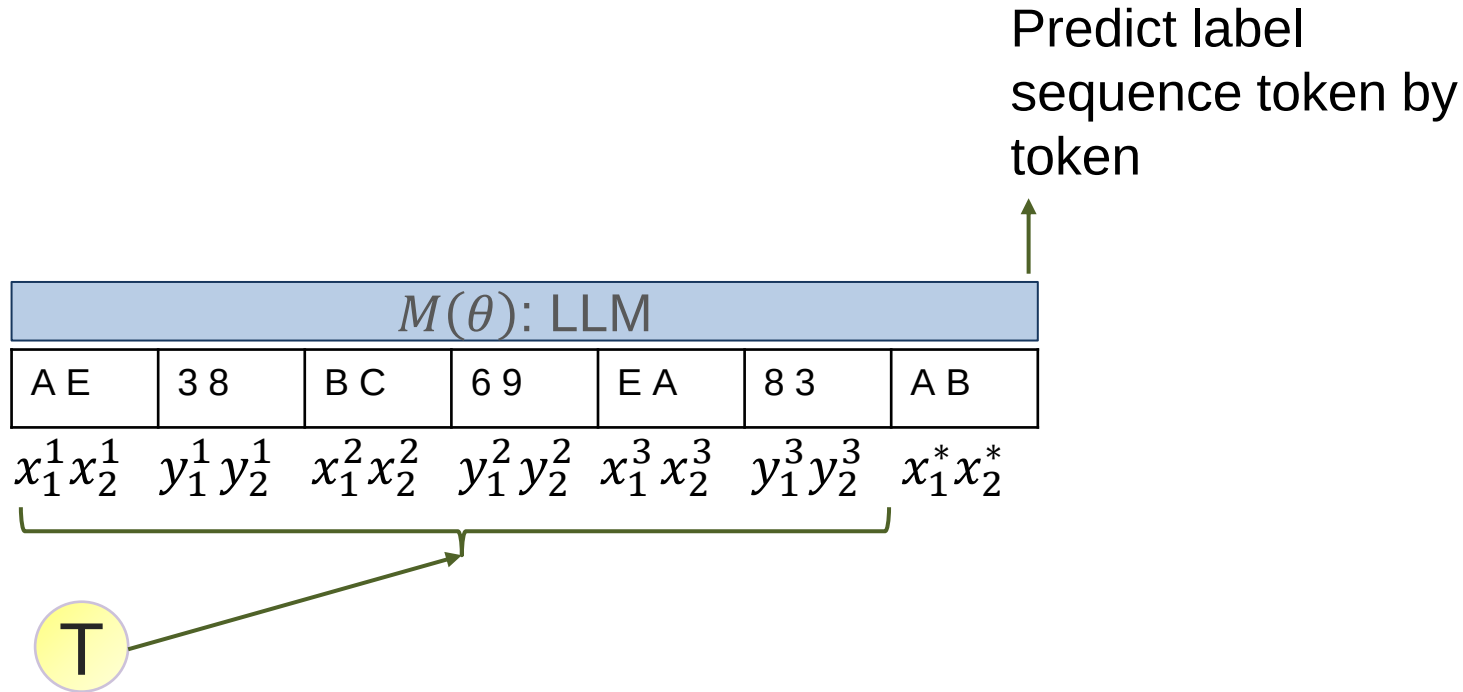
A minimal synthetic structured seq2seq task

For each task τ , fix sequence length m

- $(x_1, \dots, x_m) \sim$ Probabilistic context free grammar over alphabets
- An alignment function $A_\tau(j) \rightarrow [1..m]$.
- $(y_1, \dots, y_m \mid x_{A(1)}, \dots, x_{A(m)})$ Each y_j depends on a fixed x_i as determined by an alignment function $A_\tau(j)$
 - Special case: monotonic alignment $A(j) = j$. x and y sequence concatenation of key value pairs.

	x^1	y^1	x^2	y^2	x^3	y^3	x^*	
$m=2$	A E	3 8	B C	6 9	E A	8 3	A B	? ?
$m=3$	A B C	3 6 8	B E C	6 8 9	E A B	8 3 6	B C A	? ? ?

ICL on Sequence to Sequence tasks



The correct output $y^* = 3\ 6$ needs to be copied from parts of two previous sequences.

Think like a transformer --- a possible ideal induction circuit for monotonic alignments

11

3



Layer 3: Choose first x as query to Copy over label based on similarity of context

A	C	:	3	9	;	B	E	:	6	8	;	E	A	:	8	3	;	A	B	:
			A	C					B	E					E	A				A

Layer 2: Copy aligned x token as key using position similarity

A	C	:	3	9	;	B	E	:	6	8	;	E	A	:	8	3	;	A	B	:
1	2		1	2		1	2		1	2		1	2		1	2		1	2	

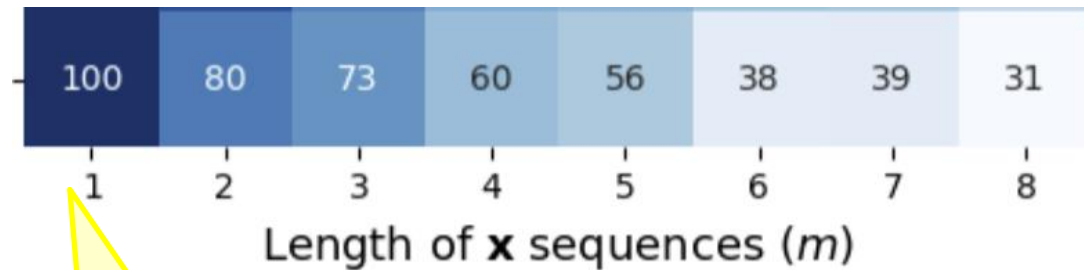
Layer 1: Context formed by assigning relative position ids to each x and y token

A	C	:	3	9	;	B	E	:	6	8	;	E	A	:	8	3	;	A	B	:
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Performance of ICL for synthetic seq2seq task

Model: LLaMA 3 with 8 billion parameters.

Accuracy drops with increasing length of the x sequence.



Basic induction
circuits formed in
Llama

Why does ICL suffer on longer sequences?

Inability to learn alignment between x, y sequences

An oracle experiment

Change y-sequence to be interleaved with aligned x-tokens.

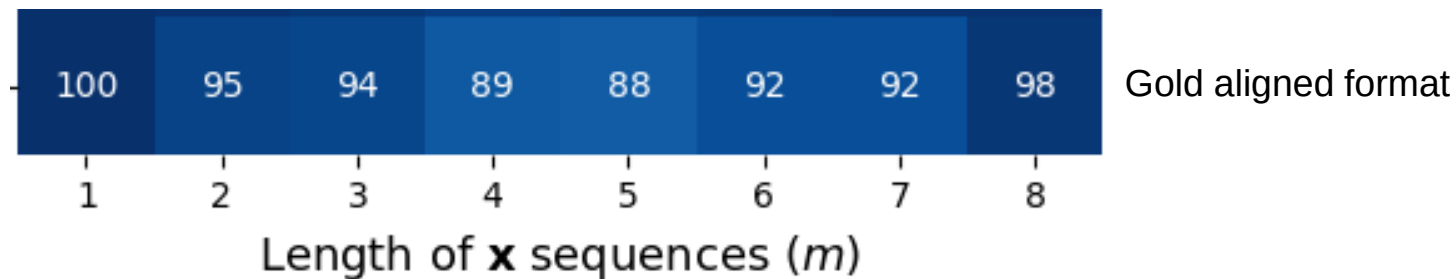
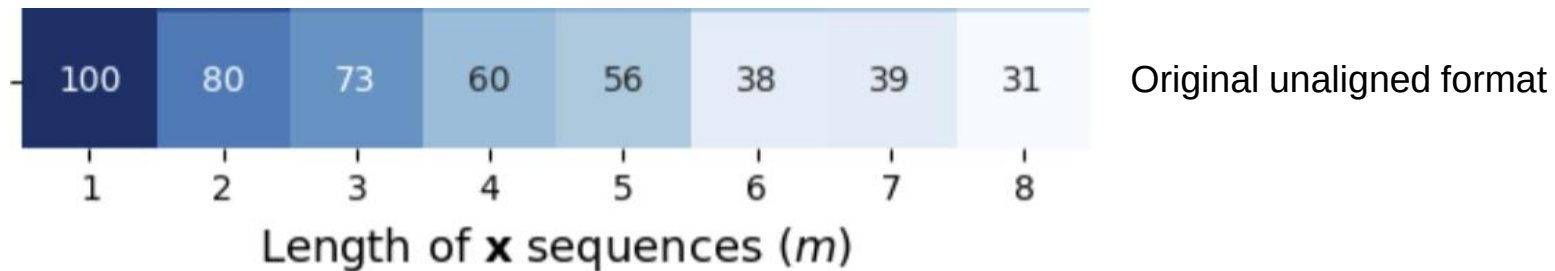
	x^1	y^1	x^2	y^2	x^3	y^3	x^*
m=3	A B C	3 6 8	B E C	6 8 9	E A B	8 3 6	A B C



Exploit oracle knowledge of alignment to interleave

	x^1	y^1	x^2	y^2	x^3	y^3	x^*
m=3	A B C	A3B6C8	B E C	B6E8C9	E A B	E8A3B6	A B C

ICL succeeds on gold aligned sequence



Can in-context learning of alignments be expressed in causal transformers?

- Seems unlikely
- Best alignment maximizes consensus in the x-y token mappings across examples
 - Classically solved via EM algorithm
 - The memory required for EM cannot be represented in causal transformers
 - Some form of chain-of-thought or scratch memory is required in the very least.

How to adapt for Seq2Seq tasks?

Full-tuning --- requires lot of training data

Alternatives:

1. Hybrid of Fine-tuning and In-context learning
2. Pre-align by decomposing structured sequence outputs
3. External pairwise matching

IC-Tune: A hybrid of in-context learning and Fine-tuning.

- Standard fine-tuning: independent loss on each example

$$\max_{\theta} \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim P_{\tau}(X, Y)} \sum_{j=1}^n \log T_{\theta}(\mathbf{x}, y_1, \dots, y_{j-1})[y_j]$$

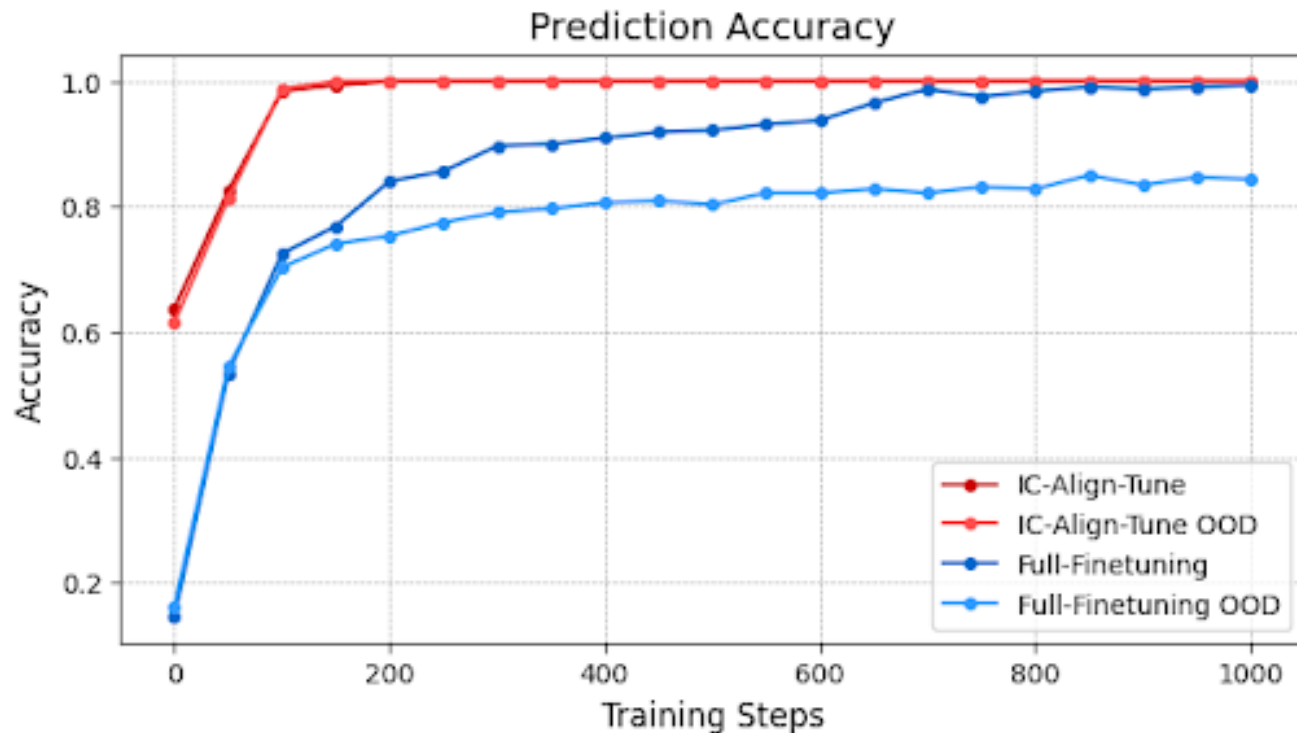
- IC-Tune: Fine-tune with in-context examples

$$\max_{\theta_A} \sum_{j=1}^n \log M_{\theta} \left(\underbrace{\mathbf{x}^1, \mathbf{y}^1, \dots, \mathbf{x}^k, \mathbf{y}^k}_{\text{In-context examples in the prompt during fine-tuning}}, \mathbf{x}^{k+1}, y_1^{k+1}, \dots, y_{j-1}^{k+1} \right) [y_j]$$

In-context examples in the prompt during fine-tuning

IC-Tune Vs Fine-tuning

IC-Tune learns faster and more robust to OODs

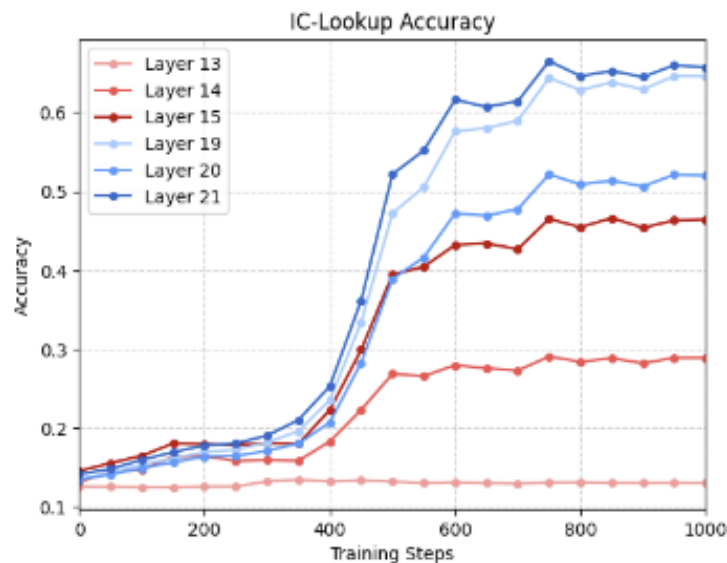
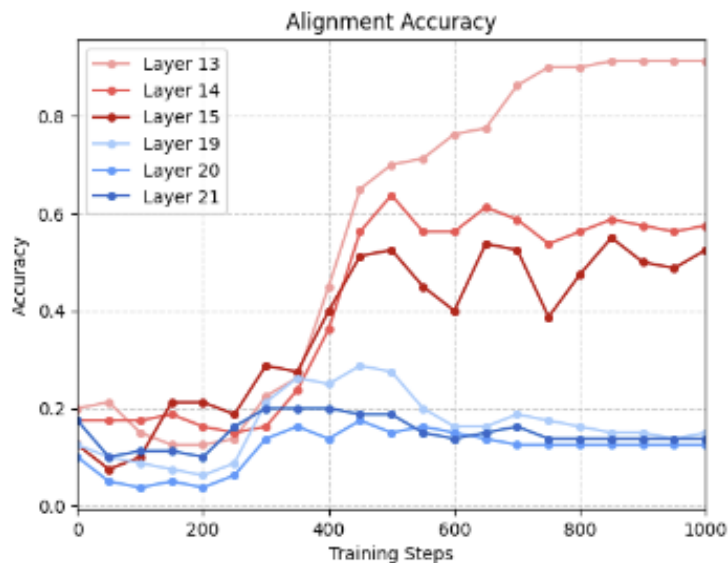


Emergence of alignment and Induction heads with IC-Tune

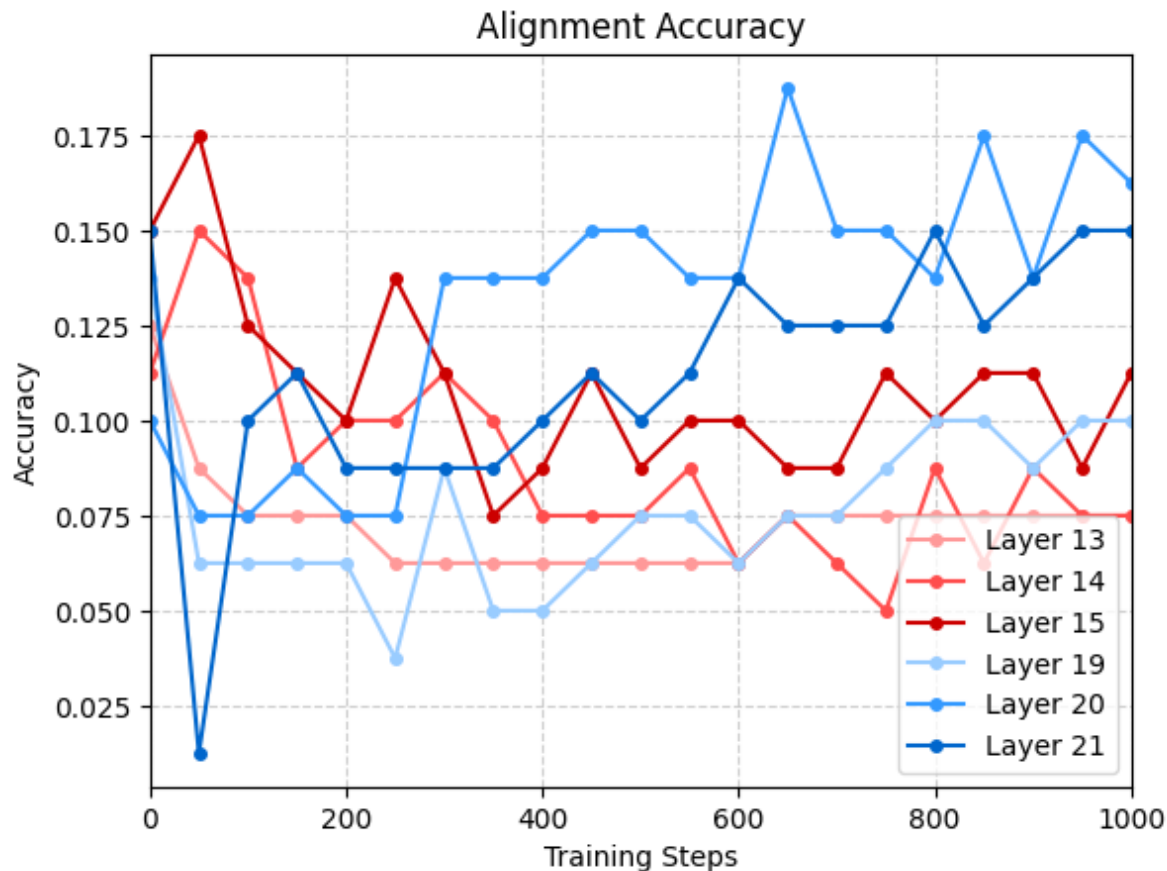
Alignment accuracy high in middle layers

Induction head matches high in upper layers

Abrupt rise in alignment accuracy, inductive lookup accuracy rises slightly later,



Fine-tuning does not uncover alignments



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1. Decompose programs into standalone subprograms

Query (x^i): Schedule a team meeting tomorrow from 3 : 00 pm to the end of the day

Code (y^i): do(Let(x0, DateTime?(date=Tomorrow(), time=NumberPM(3))) , CreateEvent(AND(has_subject(team meeting), starts_at(\$x0), ends_at(AND(GE(\$x0), EndOfWorkDay())))))

A labelled example for ICL

code: "do(Let(x0 , DateTime?(date= Tomorrow() , time=NumberPM(3))) , CreateEvent(AND(has_subject(team meeting) , starts_at(\$ x0))))"

code: "CreateEvent(AND(has_subject(team meeting) , starts_at(DateTime?(date= Tomorrow()))))",

code: "CreateEvent(has_subject(team meeting))"

code: "CreateEvent(starts_at(DateTime?(date= Tomorrow())))"

"code": "CreateEvent(starts_at(DateTime?(time= NumberPM(3))))"

code: "CreateEvent(ends_at(EndOfWorkDay()))",

1. Decompose programs into standalone subprograms
2. Attach subsuming sub-queries with each sub-program

Query (x^i): Schedule a team meeting tomorrow from 3 : 00 pm to the end of the day

Code (y^i): do(Let(x0, DateTime?(date=Tomorrow(), time=NumberPM(3))) , CreateEvent(AND(has_subject(team meeting), starts_at(\$x0), ends_at(AND(GE(\$x0), EndOfWorkDay())))))

A labelled example for ICL from T

query: "Schedule a team meeting tomorrow from 3 : 00pm",

code: "do(Let(x0 , DateTime?(date= Tomorrow() , time=NumberPM(3))) , CreateEvent(AND(has_subject(team meeting) , starts_at(\$ x0))))"

query: "Schedule a team meeting tomorrow",

code: "CreateEvent(AND(has_subject(team meeting) , starts_at(DateTime?(date= Tomorrow()))))",

query: "Schedule a team meeting",

code: "CreateEvent(has_subject(team meeting))"

query: "Schedule event tomorrow",

code: "CreateEvent(starts_at(DateTime?(date= Tomorrow())))"

"query": "Schedule event from 3 : 00 pm",

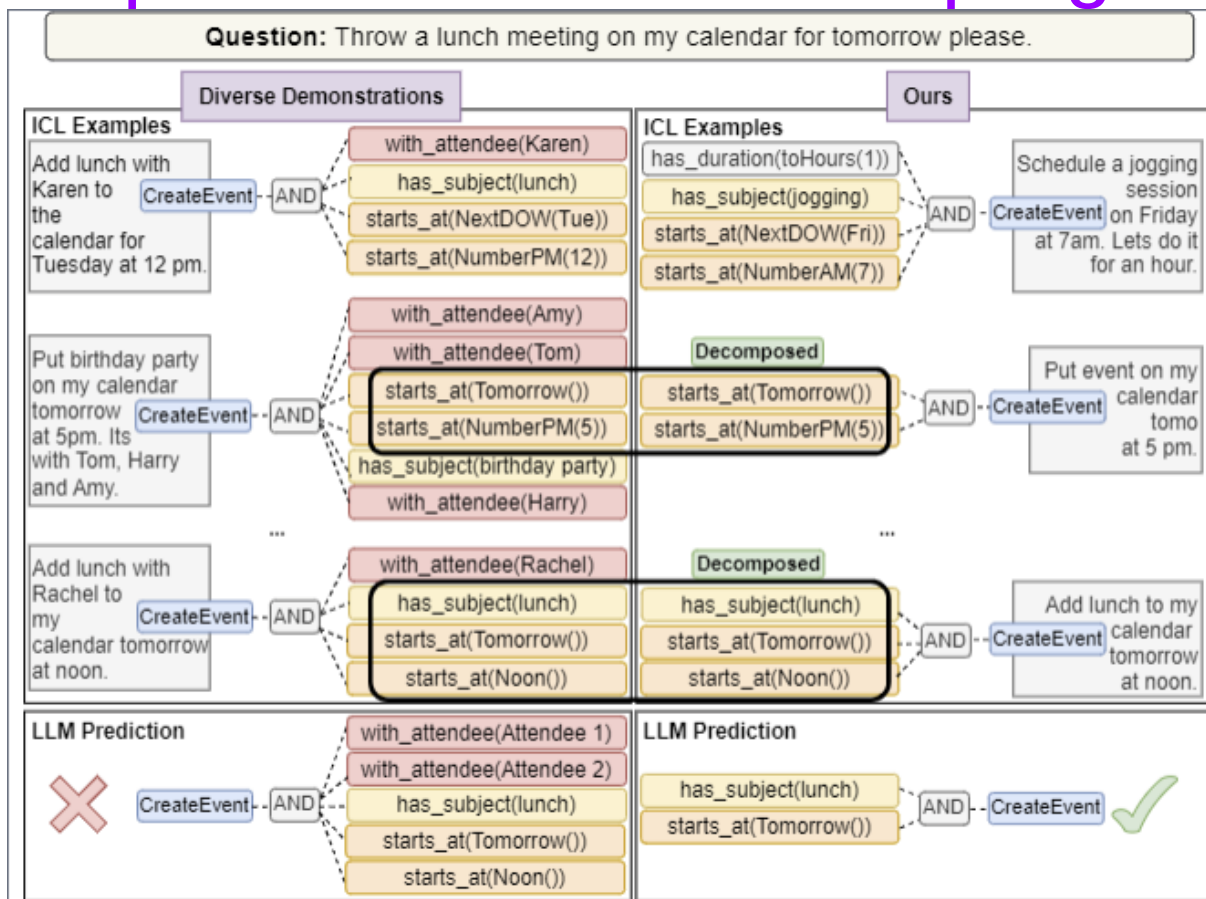
"code": "CreateEvent(starts_at(DateTime?(time= NumberPM(3))))"

query: "Schedule event to end of the day",

code: "CreateEvent(ends_at(EndOfWorkDay()))",

T augmented with smaller code-text pairs

IC examples are relevant sub-programs



Greater accuracy particularly for low-resource languages

LLM	ICL type	SMCalFlow				GeoQuery			
		EN	HI	FR	RU	length	i.i.d	template	tmcd
GPT3.5	CoverLS	42.4	43.4	40.1	48.7	50.9	77.2	86.5	70.9
	SCUD4ICL	45.8	50.2	46.0	49.5	59.2	81.9	87.7	82.0
GPT4o	CoverLS	54.1	49.6	53.5	54.5	73.6	86.3	91.6	80.7
	SCUD4ICL	54.2	51.8	54.3	55.8	81.0	86.9	89.8	85.4
Mistral:7b	CoverLS	45.8	32.7	43.5	36.5	32.3	58.9	62.3	49.4
	SCUD4ICL	46.6	37.2	46.0	39.8	50.6	67.7	74.3	73.9
Llama3:8b	CoverLS	48.2	28.5	43.5	39.3	35.6	55.8	63.4	50.9
	SCUD4ICL	46.4	34.2	43.7	40.2	44.4	52.9	58.9	48.4

Table 1: Execution accuracy on two datasets: (1) SMCalFlow with training split $\mathcal{T} = \text{T5}$ with programs in four different languages En,Hi,Fr,Ru. and (2) GeoQuery with training split $\mathcal{T} = \text{D3}$ and various test splits. All results are average of three runs and with $M = 5$. Observe that SCUD4ICL provides much higher accuracy than CoverLS in most cases, and the gains are higher for smaller LMs.

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2. Pre-align by decomposing structured sequence outputs
3. External pairwise matching

Task: ICL for Text-to-SQL generation

Database schema

x^1 : Is the molecule housing atom TR426_52 considered carcinogenic?

y^1 : SELECT T2.label AS flag_carcinogenic FROM atom AS T1 INNER JOIN
molecule AS T2 ON T1.molecule_id = T2.molecule_id WHERE
T1.atom_id = 'TR426_52'

x^2 : Is molecule TR183 known to be a carcinogen?

y^2 : SELECT T.label FROM molecule AS T WHERE T.molecule_id = 'TR183'

x^* : label='+'mean molecules are carcinogenic; Is molecule TR124
carcinogenic?

Answer:

ICL Fails on this simple case

Predicted SQL:

```
SELECT T2.label FROM atom AS T1 INNER JOIN molecule AS T2 ON  
T1.molecule_id = T2.molecule_id WHERE T1.atom_id = 'TR000_1' AND  
T2.label = '+'
```

Gold:

```
SELECT T.label FROM molecule AS T WHERE T.molecule_id = 'TR124'
```

Fails to identify highly related ICL examples that have the same SQL up to constants.

Gets distracted by random matches with constants.

Three-step fix

I: Generalize labelled examples to templates

x^2 : Is molecule TR183 known to be a carcinogen?

y^2 : SELECT T.label FROM molecule AS T WHERE T.molecule_id = 'TR183'



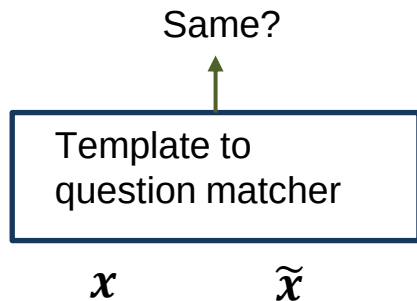
Convert to
template

$\widetilde{x}^2$: Is molecule String known to be a carcinogen?

$\widetilde{y}^2$: SELECT T.label FROM molecule AS T WHERE T.molecule_id = String

Three-step fix

II: Custom matching of test questions with templates



Repurpose a Natural language inference model for the task

Apply matcher over stored templatized labelled examples in T .

III: Template constrained decoding

If close enough template $\tilde{y}$ found, constrain generated SQL to match template

- Requires extending default LLM generation code for template constrained generation.

Encode template as a grammar

```
SELECT COUNT (T1.account_id  
.....T1.amount < Number"
```

```
SELECT WS "COUNT" WS? "(" WS?  
"T1.account_id" ..... "T1.amount" WS? "<"  
WS? "[\d]+"
```

Efficient libraries exist for grammar constrained decoding

Results

Method	Bird-Dev		Spider-Dev	
	CodeS-1B	Llama-3.1	CodeS-1B	Llama-3.1
ZeroShot	40.91%	44.81%	70.33%	77.01%
ICL-3	50.65%	73.90%	79.37%	93.52%
SGC	55.20%	84.15%	82.32%	95.09%
TeCoD	82.60%	87.92%	97.45%	97.64%

Workload: Every query has a gold template in the labelled pool

Template constrained decoding more accurate than ICL
Small 1B models comparable to larger models.

Visit our poster: “Template Constrained Text-to-SQL Generation for Frequent Queries”

Concluding Remarks on ICL

- ICL is convenient for on-the-fly adaptation in multi-tenancy models.
 - Accuracy: decent
 - Robustness: requires improvement, on the mend.
 - Efficiency: High, can be improved with KV cache
- More work needed to
 - Understand the mechanism behind ICL

Concluding remarks

- ICL can learn totally unseen tasks for scalar classification and Markov languages using induction heads
- ICL fails for unseen seq2seq learning tasks because input/output alignments cannot be learned in-context.
- A hybrid of ICL and fine-tuning can be harnessed to only learn the missing alignment function.

Concluding remarks

- When fine-tuning is not an option
 - Decomposition of large structured outputs may be useful
 - Template-driven representation may increase overlap
 - External matcher to ascertain matched templates and enforce with grammar constrained decoding much better than ICL.

Open questions: ICL for Few-shot adaptation

- Formally establish the representation capability of the transformer to in-context learn various tasks
- Can an additional scratch memory or chain-of-thought significantly enhance its capability?
- What form of pre-training can cause the emergence of ICL over structured tasks such as seq2seq learning?
- Better harnessing of long-context LLMs for robust and efficient ICL over large number of examples

Thank you all for listening